

A Software-Agnostic Small Studio and Education Pipeline

Zach Wood
zach.wood13@gmail.com
Brigham Young University
United States

Anna Hales
a.hales579@gmail.com
Brigham Young University
United States

Matthew Minson
matthew.minson.912@gmail.com
Brigham Young University
United States

Seth Holladay
seth_holladay@byu.edu
Brigham Young University
United States

Craig Van Dyke
craig@cs.byu.edu
Brigham Young University
United States



Figure 1: Layout in Maya, Unreal, and Renderman in Houdini (left to right) all referencing the same USD file.

ABSTRACT

Building a production pipeline from scratch that supports iteration and creative control, for a university or small studio, can be difficult to achieve with smaller budgets and major time constraints. We developed a USD pipeline that allows our production to work seamlessly between multiple production software packages, playing to the strengths of each. With these tools, we were able to build our entire animation pipeline in just months and adapt it on the fly to the needs of our production. This pipeline also allowed artists to choose different software based on the strength of each package, all while referencing the same asset and layout files.

ACM Reference Format:

Zach Wood, Anna Hales, Matthew Minson, Seth Holladay, and Craig Van Dyke. 2023. A Software-Agnostic Small Studio and Education Pipeline. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Educator's Forum (SIGGRAPH '23 Educator's Forum)*, August 06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3587424.3595578>

1 INTRODUCTION

Building a production pipeline from scratch that supports iteration and creative control can be an expensive and difficult task, especially for a university or small studio limited by budget and timeline. Different software packages provide different tools to produce various parts of an animation, and creating a pipeline with a small team that coordinates work between packages to handle complex

stories and worlds can require a significant amount of overhead. USD is a powerful solution that simplifies workflows and makes it possible for artists to work in an iterative, art-directable way, without spending unnecessary time dealing with the challenges of working across different packages. Our small team of undergraduate students managed to build an entire flexible pipeline around USD and Solaris tools in months, designed to mirror professional studio production and supporting *all* teams across the pipeline with their tools of preference.

Examples of the interesting problems and complexities of our production include:

- **Complex layout** with hundreds of assets to be organized, referenced, and updated simultaneously across Houdini, Maya, and Unreal with shot-based and sequence-based layout edits. Story demanded continuous changes to asset configurations.
- **High-quality previs cut** with preview surface shaders and real-time FX and lighting for real-time iteration.

2 PIPELINE OVERVIEW

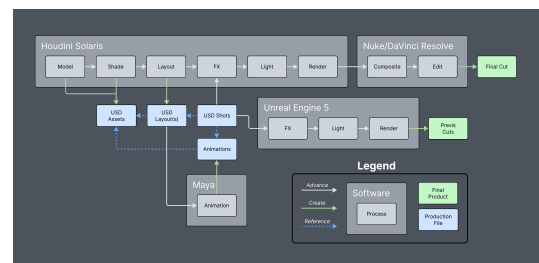


Figure 2: An overview flowchart of our pipeline.

Houdini was our central software package, with our workflow based around Solaris' USD tools. We modeled assets in Maya, then

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH '23 Educator's Forum, August 06–10, 2023, Los Angeles, CA, USA

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0146-7/23/08.

<https://doi.org/10.1145/3587424.3595578>

imported them into Houdini for layout, shading, FX, and rendering. We created an automated script in Houdini to build layered asset USDs referencing their geometry and materials, then began building set layouts in Houdini, using the pre-visualization versions of the assets. Layout artists and animators tweaked those same layouts in Maya for previs and shot composition, including edits on both a shot-based and sequence-based level. Our rigging, animation, and previs teams also rigged, animated, and exported cameras and character animation from Maya as alembics, which were then brought into Houdini and re-exported with a USD wrapper around the alembic, meaning that every element in a scene – from sets to character animation to camera – was in USD format. In Unreal, artists read in the USD files and create previs versions of the shots to reveal needed modifications. Our FX and lighting teams were also able to work and iterate in real time during the previs process, allowing them to speed up their look development process and to solve problems earlier in the production process. In Houdini, all elements of a shot were brought back together for final layout, FX, lighting, and rendering. In this way, USD provided a straightforward framework which allowed us to transfer the same files and data between all of these industry standard software packages, using the strengths of each program in an artist-friendly, iterative workflow.

With the use of USD, modifications at any level were automatically propagated up and down the pipeline, allowing for constant iteration and development without the need for extra tools or overhead.

3 ASSETS

Production was further expedited by USD's "purpose" functionality. Purposes define multiple sets of geometry and materials within a USD asset, with the geo/material pair being loaded depending on the environment in which the file is accessed. A USD file with defined purposes can be accessed and rendered appropriately in any package in our pipeline, without artists needing to worry about loading the proper version of each asset themselves.

The main hurdle that purposes helped us overcome in our pipeline was the varying shader specifications used by different programs. In the past, these discrepancies meant materials had to be completely separate from their corresponding geometry, forcing artists to rebuild each asset's material if they wanted to view it in another package, or to simply do without materials until render time. With purposes, this was no longer the case – each asset's material could be embedded in its USD file. Thus, a single USD could be loaded in any number of software packages, with each package displaying the asset fully shaded without additional steps, streamlining artists' workflows and simplifying the pipeline significantly.

4 PREVIS WORKFLOW

Being able to work across multiple programs aided look development significantly. Previs artists could work in Maya to block in camera work in a set made up of proxy models/materials. This allowed the previs department to move forward while modeling and shading were taking place, instead of being forced to wait for higher-quality, finalized assets to begin their work. Then, as artists created final geometry/materials, they could easily update the assets and save the updated version to the USD asset on disk. Once

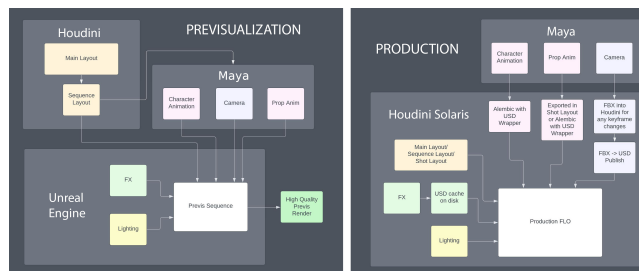


Figure 3: A depiction of the previs vs. production workflows.

an artist updated an asset, it became immediately accessible by all other artists on the film in Unreal, Maya, and Houdini.

5 FINAL SHOT SETUP AND RENDERING

For final shot setup, all elements of a scene were referenced into a Houdini file, where lighting, FX, and set dressing for that shot were polished. Although animation was done in Maya and final render took place in Houdini, animators could easily make shot-based changes to be applied in all packages because both Maya and Houdini referenced the same USD files.

Another critical aspect of our pipeline was a standardized set of custom HDAs that formed the backbone for every shot. These HDAs were wrappers around LOP networks, performing various tasks such as bringing in the appropriate layout, cameras, and animation for any given shot. They also provided workspaces for FX artists to develop, cache, and reference character and environment effects, which could then be seamlessly integrated into the scene for lighting and final render. Providing these tools in custom nodes made setting up and organizing shots much quicker and simpler. It also allowed TDs to update the HDAs as needed throughout production, then to simply update the node's definition to make the updates available in every shot in the film.

In addition to improving our process in asset-based and shot-based departments, our USD-based pipeline also greatly helped our rendering team. With previous pipelines, creating scene description files was one of the most expensive parts of the rendering process. Before a shot could be sent to the render farm, a scene description file had to be generated for each frame. This process was extremely time-consuming, could not be farmed out, and required large amounts of disk space to store all the per-frame scene description files.

However, our updated pipeline allowed us to generate a single reference-based USD scene description file for an entire shot (a relatively quick process). We could then use Husk (Houdini's command line USD rendering tool) to send jobs to our render farm. Not only was this process much quicker, it was also much more storage-efficient, only requiring a single scene description file to be stored on disk for each shot.

6 RESULTS

All of these elements combined to create a smooth, consistent workflow that benefited all teams and shots. The pipeline came together in about 4 months, built from the ground up, after which we finished 123 shots in 6 months.